EV Component Database

Your Complete Reference for Electric Vehicle Components

Overview

The JC Tech EV Component Database is a meticulously organized catalog of every critical component in modern electric vehicles. Designed for automotive engineers, EV startups, procurement teams, and students, this database provides specifications, supplier information, design considerations, and comparative data for informed decision-making in EV development.

Component Categories

- Battery Systems: Li-ion cells (NMC, LFP, NCA), battery packs, BMS (Battery Management System), thermal management modules, and cell balancing circuits.
- Electric Motors: BLDC, PMSM, induction motors, switched reluctance motors -- torque curves, efficiency maps, and cooling strategies.
- Power Electronics: DC-DC converters, onboard chargers (OBC), inverters (IGBT/SiC/GaN), and high-voltage distribution units (PDU).
- Charging Infrastructure: AC Level 1/2, DC fast charging (CCS, CHAdeMO, GB/T), wireless charging pads, and V2G (Vehicle-to-Grid) systems.
- Drivetrain & Transmission: Single-speed reducers, multi-speed transmissions, differential units, and half-shaft assemblies for EV applications.
- Auxiliary Systems: Electric power steering (EPS), electric AC compressors, 12V accessory batteries, and regenerative braking controllers.

Data Fields Per Component

Specifications:	Voltage, current, power rating, weight, dimensions, operating temperature range.
Suppliers:	OEM and tier-1 supplier list with region availability and lead times.

Standards:	Compliance info -- AIS 156, ISO 6469, IEC 61851, SAE J1772, UN ECE R100.
Cost Estimate:	Approximate unit pricing at volume tiers (100, 1K, 10K units).

Sample Entry: Lithium Iron Phosphate (LFP) Cell

Chemistry:	LiFePO4 (LFP)
Nominal Voltage:	3.2V per cell
Energy Density:	90-160 Wh/kg
Cycle Life:	2000-5000 cycles at 80% DoD
Thermal Runaway Risk:	Very Low (inherently stable cathode)
Common Suppliers:	CATL, BYD, EVE Energy, Gotion High-Tech
Applications:	Electric buses, commercial EVs, energy storage systems

How Engineers Use This Database

1. Component Selection: Compare multiple components side-by-side using spec sheets.
2. Design Validation: Cross-reference component specs with vehicle-level requirements.
3. Supplier Scouting: Identify and evaluate suppliers for procurement negotiations.
4. Learning & Training: Students and junior engineers use entries as study material for EV systems.

Database Growth & Roadmap

- Current: 350+ components across 6 major categories.
- Q1 2027: Adding fuel cell components and hydrogen storage systems.
- Q2 2027: Interactive comparison tool on the JC Tech web platform.
- Ongoing: Community contributions from EV engineers worldwide.

Get Involved

The EV Component Database is a living resource. If you are a component manufacturer or an EV engineer with expertise, we invite you to contribute data and reviews. Contact us to become a knowledge partner.

Visit: www.jctech.co.in | Email: ev@jctech.co.in